

E.KEERTHANA

Btch:51

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ASSIGNMENT-10.2

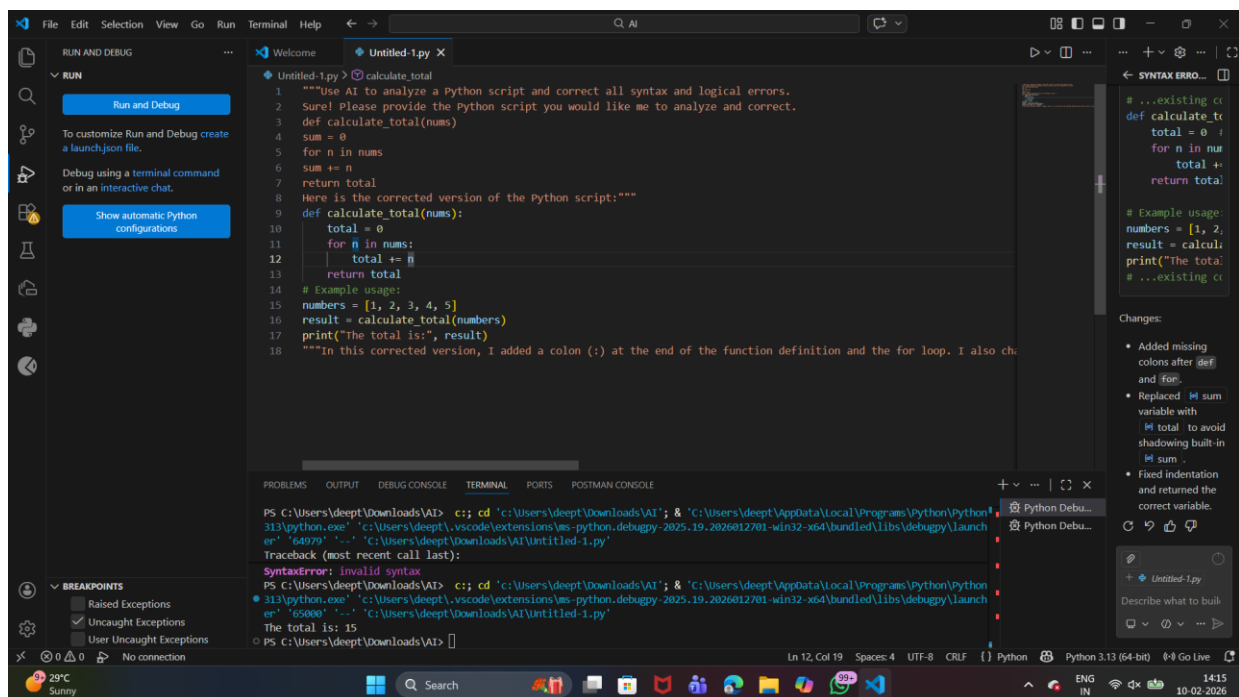
Code Review and Quality: Using AI to Improve Code

Quality and Readability

Task Description -1(Error Detection and Correction)

Prompt: Analyze the given Python code, identify all syntax and logical errors, and provide a corrected executable version with brief explanations

AI-Generated Code:

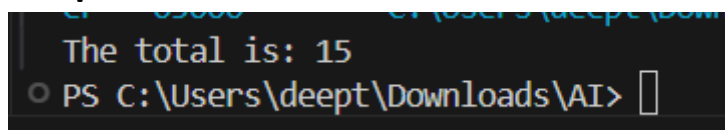


```
1 """Use AI to analyze a Python script and correct all syntax and logical errors.
2 Sure! Please provide the Python script you would like me to analyze and correct.
3 def calculate_total(nums)
4 sum = 0
5 for n in nums
6 sum += n
7 return total
8 Here is the corrected version of the Python script:'''
9 def calculate_total(nums):
10     total = 0
11     for n in nums:
12         total += n
13     return total
14 # Example usage:
15 numbers = [1, 2, 3, 4, 5]
16 result = calculate_total(numbers)
17 print("The total is:", result)
18 """In this corrected version, I added a colon (:) at the end of the function definition and the for loop. I also changed the indentation of the loop body to match the function definition's indentation.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\deep\Downloads\AI> cd 'c:\Users\deep\Downloads\AI'; & 'c:\Users\deep\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\deep\.vscode\extensions\ms-python.debugpy-2025.19.2026012701-win32-x64\bundle\libs\debugpy\launch'
er' '64979' '-.' 'c:\Users\deep\Downloads\AI\Untitled-1.py'
Traceback (most recent call last):
  File "c:\Users\deep\Downloads\AI\Untitled-1.py", line 12, in calculate_total
    total += n
  File "c:\Users\deep\Downloads\AI\Untitled-1.py", line 12, in calculate_total
    total += n
SyntaxError: invalid syntax
PS C:\Users\deep\Downloads\AI> cd 'c:\Users\deep\Downloads\AI'; & 'c:\Users\deep\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\deep\.vscode\extensions\ms-python.debugpy-2025.19.2026012701-win32-x64\bundle\libs\debugpy\launch'
er' '65000' '-.' 'c:\Users\deep\Downloads\AI\Untitled-1.py'
The total is: 15
PS C:\Users\deep\Downloads\AI>
```

Output:



```
The total is: 15
PS C:\Users\deep\Downloads\AI>
```

Justification:

Missing colon after function definition → Added :

Missing colon in for loop → Added :

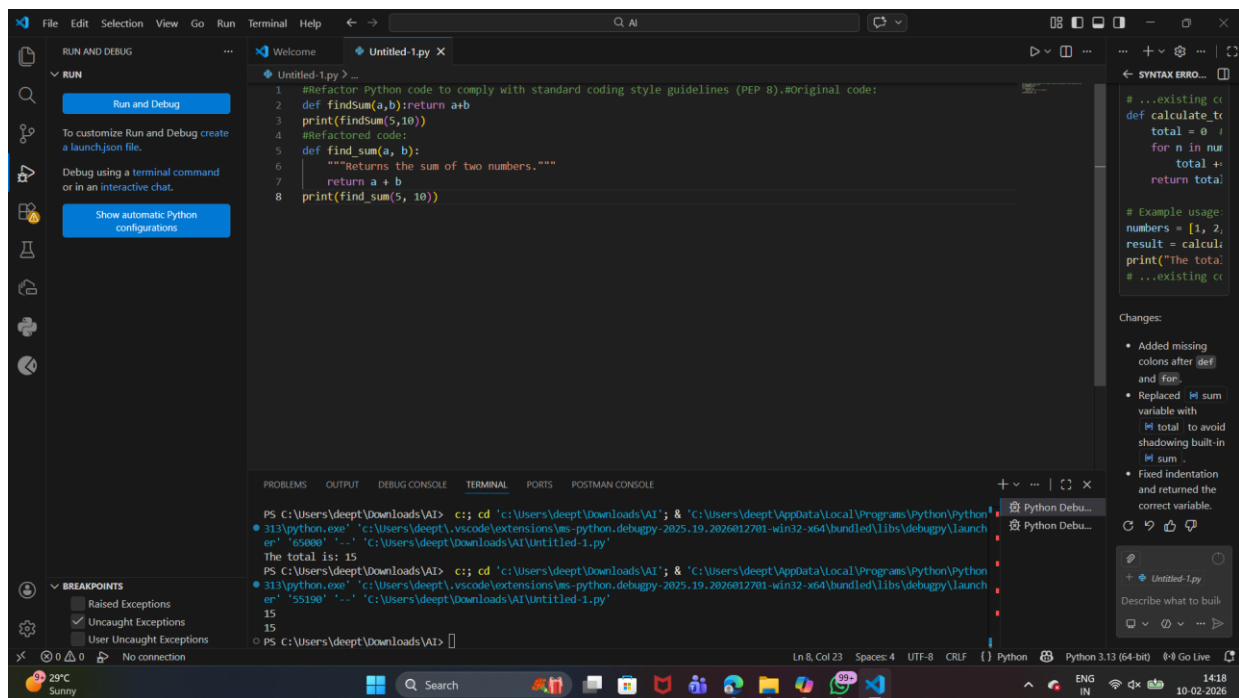
Used sum as variable name (overwrites built-in) → Renamed to total

Returned total though variable was named sum → Corrected consistency

Task Description -2(Code Style Standardization)

Prompt : Refactor the Python code to follow PEP 8 coding standards with proper formatting and readability

AI-Generated Code:



The screenshot shows the Visual Studio Code interface with a Python file named 'Untitled-1.py'. The code is being refactored to follow PEP 8 standards. The original code (lines 1-3) defines a function `findSum(a,b)` that returns `a+b` and calls it with `print(findSum(5,10))`. The refactored code (lines 4-8) defines a function `find_sum(a, b)` with a docstring, returns `a + b`, and calls it with `print(find_sum(5, 10))`. The right sidebar shows a 'SYNTAX ERROR' panel with a list of changes: 'Added missing colons after def and for.', 'Replaced sum variable with total to avoid shadowing built-in sum.', and 'Fixed indentation and returned the correct variable.' The bottom status bar indicates 'Ln 8, Col 23, Spaces: 4, UTF-8, CRLF, Python, Python 3.13 (64-bit)'.

```
1 #Refactor Python code to comply with standard coding style guidelines (PEP 8).#Original code:
2 def findSum(a,b):return a+b
3 print(findSum(5,10))
4 #Refactored code:
5 def find_sum(a, b):
6     """Returns the sum of two numbers."""
7     return a + b
8 print(find_sum(5, 10))
```

Changes:

- Added missing colons after `def` and `for`.
- Replaced `sum` variable with `total` to avoid shadowing built-in `sum`.
- Fixed indentation and returned the correct variable.

OUTPUT:

```
er' '55190' '--' 'C:\Users\deept\Downloads\AI\  
15  
15  
PS C:\Users\deept\Downloads\AI> 
```

Justification:

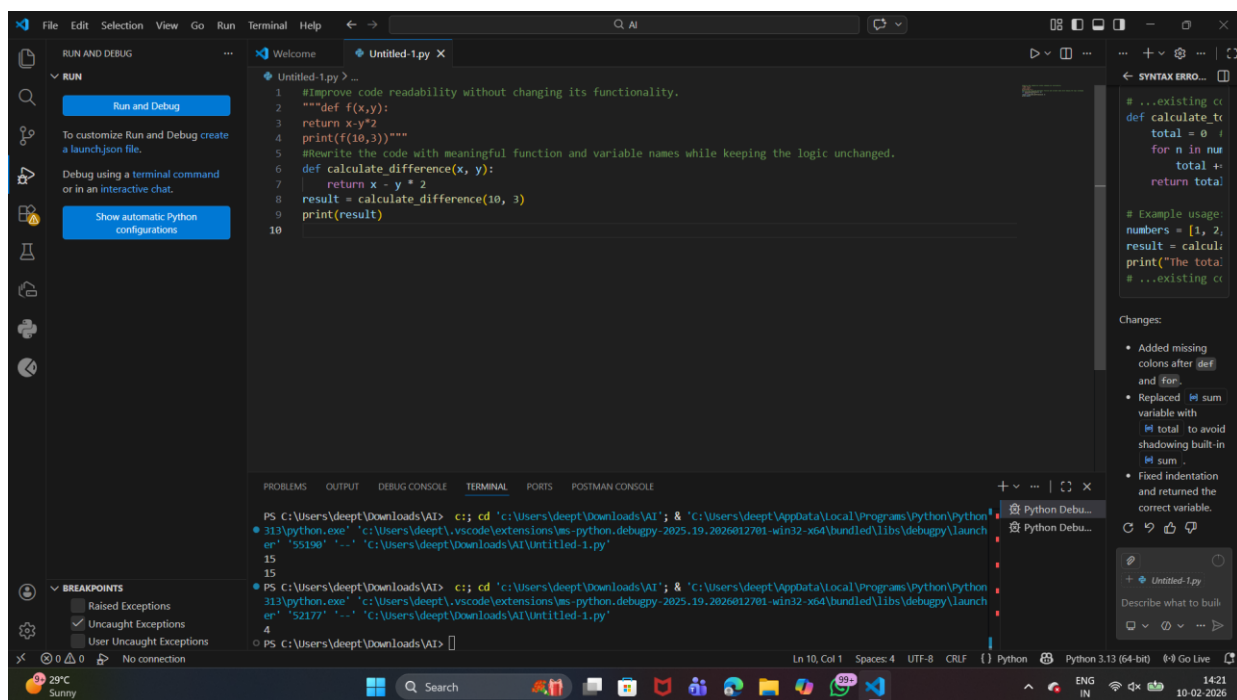
Added proper indentation and spacing

Used snake_case naming (PEP-8 standard)

Split single-line function into readable multi-line format

Task Description -3(Code Clarity Improvement)

Prompt: Rewrite the code with meaningful function and variable names while keeping the logic unchanged.



OUTPUT:

```
313\python.exe' 'c:\Users\deept\.vscode\extensions\ms-python.debug
er' '52177' '--' 'C:\Users\deept\Downloads\AI\Untitled-1.py'
4
PS C:\Users\deept\Downloads\AI> 
```

Justification:

Meaningful function and variable names

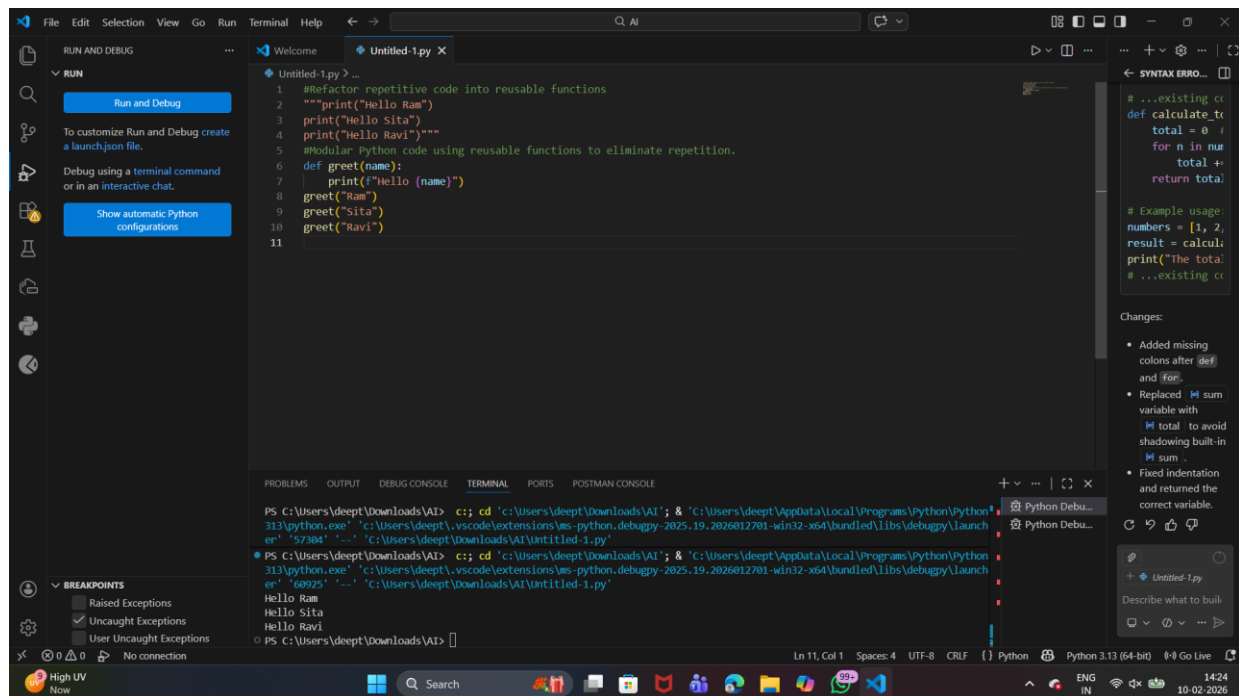
Docstring for clarity

Proper indentation and spacing

Task Description -4(Structural Refactoring)

Prompt: Refactor the repeated print statements into a reusable function.

AI-Generated Code:



OUTPUT:

```
er' '60925' '--' 'C:\Users\deepthi\Downloads\AI\Untitled-1.py'
Hello Ram
Hello Sita
Hello Ravi
PS C:\Users\deepthi\Downloads\AI>
```

Justification:

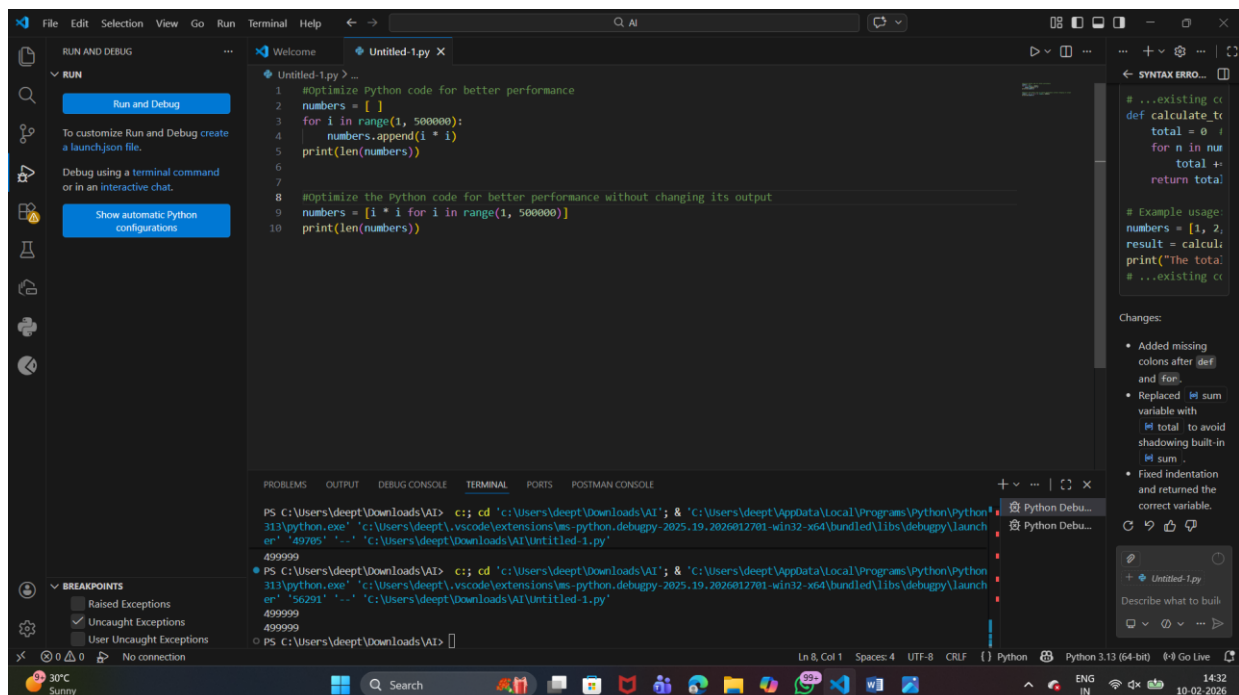
Eliminated repetition

Created reusable function greet()

Task Description -5(Efficiency Enhancement):

Prompt: Optimize the Python code for better performance without changing its output.

AI-Generated Code:



OUTPUT:

```
515 (python.exe) C:\Users\deept\VSCode\extensions\ms-py...  
er' '56291' '--' 'C:\Users\deept\Downloads\AI\Untitled-1'  
499999  
499999  
PS C:\Users\deept\Downloads\AI> █
```

Justification:

- Performance Improvements
- Used list comprehension instead of loop
- Faster execution
- Cleaner and more Pythonic code